Daniil Yuzepenko

JUNIOR GO (GOLANG) DEVELOPER | SOFTWARE ENGINEER

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> Junior Go Developer and 3rd-year Computer Science student with practical commercial experience building high-performance microservices. Possesses deep knowledge of Go runtime internals (compiler AST/SSA phases, Escape Analysis memory graph optimization, GMP scheduler mapped to OS threads/CPU cores, channels, and map memory layouts). Proficient in high-throughput backend architecture, gRPC/REST APIs, and robust network security protocols (TCP handshake, OSI, TLS/mTLS). Former Go Intern at EPAM Systems with proven ability to engineer production-ready, clean, and scalable services.

TECHNICAL DEEP-DIVE & SKILLS

GO RUNTIME & INTERNALS

Escape Analysis & Memory: AST compilation phases, AST escape graph analysis determining stack vs. heap allocation.

GMP Scheduler: Goroutine state transitions, OS thread binding (m0/g0), Processor local runqueues, work-stealing, sysmon preemption, CPU core affinity.

Channels & Concurrency: Internal hchan struct, basic understanding of direct stack copy via go_direct(), goroutine parking/unparking via go_park() and goready().

Map Architecture: Hash indexing using LOB (Low-Order Bits for bucket indexing) and HOB (High-Order Bits for hash matching within bucket slots), 8-key/value array bucket layout, overflow bucket chains, understanding memory reallocation.

NETWORKING & SECURITY

Network Stack: OSI 7-Layer model, TCP protocol state machine, 3-way handshake, SYN/ACK, TCP.

TLS/mTLS Security: Asymmetric key exchange (RSA/ECDHE), Symmetric payload encryption (AES-GCM), Certificate Authority (CA) validation, dual-way client certificate authentication (mTLS).

CLOUD INFRASTRUCTURE, DOCKER & KUBERNETES

Containerization & Optimization: multi-stage Docker builds for Go binaries, layer caching optimization.

Kubernetes Architecture: Solid grasp of K8s internals, including Control Plane components (api-server, etcd, scheduler, controller-manager), Worker Node components (kubelet, kube-proxy), and workload lifecycle management (Deployments, StatefulSets, DaemonSets, Jobs/CronJobs).

Infrastructure as Code (IaC): Terraform state management, Provider-based cloud resource provisioning, and automated CI/CD pipelines via GitHub Actions.

Backend & Architecture: Go (Golang), gRPC, Protocol Buffers, grpc-gateway, REST API, Gin, GORM, JSON, WebSockets, Clean Architecture, Event-Driven Architecture, SOLID, DRY.

Databases & DevOps: PostgreSQL, Redis (caching patterns, TTL), SQL/NoSQL, Docker, Docker Compose, Kubernetes, Nginx, CI/CD (GitHub Actions), VPS deployment, Git.

PROFESSIONAL EXPERIENCE

EPAM Systems — Go Developer Intern

May 2026 – Jun 2026

Project: EPMT-SPGO | Enterprise Project Tracking Microservice

- Architected and implemented a high-concurrency gRPC microservice in Go delivering complete CRUD and List operations for Users, Issues, and Projects.
- Designed language-agnostic Protocol Buffer contracts, exposing dual gRPC and HTTP/REST JSON endpoints via grpc-gateway.
- Applied Clean Architecture and SOLID principles to decouple core domain logic from transport layers and database implementations.
- Implemented memory-efficient data structures by leveraging compiler knowledge and zero-allocation techniques during request parsing and validation.
- Engineered a dynamic storage abstraction layer supporting interchangeable in-memory, PostgreSQL, and Redis caching implementations with configurable TTL.
- Containerized service stack with Docker Compose, configuring persistent volumes, network isolation, and health checks. Passed final technical assessment confirming production readiness.

Intechnity — **Junior Developer** Opole, Poland

- Developed and maintained performant back-end web services and APIs using C# and Entity Framework Core.
- Wrote comprehensive integration test suites and optimized data access patterns across Repository and HTTP handler layers.
- Collaborated closely with cross-functional teams on React-based front-end integration and API consumption.

SELECTED HIGH-PERFORMANCE PROJECTS

JoyJoin (Event Management Multi-Tool Platform) Backend & DevOps Engineer

- Co-architected a high-concurrency event management platform in Go, engineered to handle heavy concurrent user registration bursts.
- Leveraged low-level concurrency primitives (Worker Pools, Semaphores, WaitGroups) alongside goroutine channel tuning (go_park aware) to prevent resource starvation under load.
- Configured secure microservice-to-microservice communication incorporating mTLS handshakes and Sidecar proxies for encrypted service mesh communication.
- Managed full deployment pipeline using Docker, Nginx reverse proxying, and automated GitHub Actions CI/CD workflows.

Doodle Jump Motion Online Full-Stack & DevOps Developer

- Engineered a real-time multiplayer game platform featuring camera-based motion controls using MediaPipe.
- Built a low-latency WebSockets communication server in Go for bidirectional state synchronization and dynamic player matchmaking.
- Set up secure SSL/TLS termination, Nginx reverse proxying, containerization via Docker, and continuous integration to a production VPS.

EDUCATION

Opole University of Technology — **Bachelor of Engineering in Computer Science** Expected 2027 (Currently 3rd Year)

Focus on Computer Networks, Operating Systems, Systems Programming, Algorithms & Data Structures, and Software Engineering.

LANGUAGES

Ukrainian: Native **Polish:** Fluent **English:** Upper-Intermediate (B2)